

ALHASSAN AHMED

Data Scientist | Machine Learning Engineer

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PROFESSIONAL SUMMARY

Data Science and Machine Learning graduate (B.Sc. Computer Science, 2026) who builds and ships end-to-end AI systems: data preprocessing, predictive modeling, NLP and computer vision, LLM/RAG applications, and on-device deployment via ONNX Runtime, FastAPI, and Docker. Delivered a wearable health-monitoring platform combining a 0.971 macro-AUC fall-detection CNN, a 64 KB on-device cardiac classifier, and a bilingual RAG clinical assistant that passed a 64-prompt safety evaluation at 100%. Ready to convert raw sensor, financial, and text data into deployed, measurable products.

TECHNICAL SKILLS

Programming Languages: Python (OOP, Data Structures), SQL, Java, C++, C#, JavaScript, TypeScript, Kotlin

Machine Learning & Deep Learning: Scikit-learn, PyTorch, TensorFlow, Keras, CNNs, ANNs, Predictive Modeling, Classification, Regression, Clustering, Knowledge Distillation, Model Evaluation (Precision, Recall, F1-Score, ROC-AUC), Imbalanced Data Handling

NLP & LLMs: Natural Language Processing, TF-IDF, Sentiment Analysis, Large Language Models (Llama 3.3 70B), Retrieval-Augmented Generation (RAG), FAISS Vector Search, Prompt Engineering, LLM Safety (Llama Guard)

Data Analysis & Visualization: Pandas, NumPy, Data Preprocessing, Feature Engineering, Exploratory Data Analysis (EDA), Matplotlib, Seaborn, Power BI, Interactive Dashboards

Deployment & MLOps: FastAPI, Flask, RESTful APIs, Docker, ONNX Runtime (on-device inference), Hugging Face Spaces, Git, GitHub

Databases & Backend: SQL Server, MongoDB, PostgreSQL (Supabase), ASP.NET Core MVC, Entity Framework Core

PROJECTS

Smart Health Monitoring System — Graduation Project

Future University in Egypt | Spring 2026

End-to-end AI health platform for elderly and chronically ill users: on-device deep learning for fall detection, activity recognition, and ECG classification, paired with an LLM/RAG clinical assistant, wearable sensor integration, geofencing, and automatic emergency escalation.

- Achieved 0.971 macro-AUC fall detection with a ~3.5% false-positive rate, validated under 9-fold Leave-One-Subject-Out evaluation, by designing FusionNet, a dual-stream 1D-CNN trained on wearable accelerometer and gyroscope data.
- Reached 94.4% per-window activity-recognition accuracy (95.9% with bout voting) on the WISDM benchmark using a subject-independent split by building and tuning a dual-head 1D-CNN.
- Compressed a cardiac beat classifier to a 64 KB on-device model with 0.96 held-out Lead-I N-recall by applying knowledge distillation (Student-CNN) from a larger teacher network.
- Scored a 100% pass rate on a 64-prompt clinical safety evaluation by engineering a bilingual (English/Arabic) RAG assistant with Llama 3.3 70B, FAISS retrieval over a curated medical corpus, a deterministic rules engine, and drug-interaction and red-flag detection.
- Enabled fully offline, privacy-preserving inference at sub-millisecond to sub-10 ms latency by exporting all three deep learning models to ONNX Runtime, keeping raw sensor data on the device.
- Validated the full alert-and-escalation pipeline on real hardware (Samsung Galaxy Watch 5), including a staged physical fall that triggered caregiver alerts, SOS, and emergency escalation end to end.
- Hardened the system with a defense-in-depth safety architecture, combining deterministic emergency flagging with graceful degradation under LLM-provider failures.

Tech Stack: Python, PyTorch, TensorFlow/Keras, ONNX Runtime, FAISS, Groq (Llama 3.3 70B), Llama Guard, FastAPI, Supabase (PostgreSQL), React Native, Expo, TypeScript, Kotlin (Wear OS), Docker, Hugging Face Spaces

Credit Risk & Loan Approval Prediction

- Predicted loan approval outcomes and quantified credit risk from customer financial data by engineering features, encoding categorical variables, and benchmarking Scikit-learn classifiers with ROC-AUC and F1-score.

Tech Stack: Python, Pandas, Scikit-learn

Fraud Detection System

- Flagged fraudulent transactions on a highly imbalanced dataset by applying resampling strategies and optimizing the precision-recall trade-off to minimize missed fraud cases.

Tech Stack: Python, Scikit-learn

Customer Churn Prediction

- Identified customers at risk of churning from behavioral and transactional data by training and comparing classification models, enabling targeted retention decisions.

Tech Stack: *Python, Pandas, Scikit-learn*

NLP-Based Customer Feedback Analysis

- Extracted sentiment and recurring complaint themes from customer reviews by building an NLP pipeline with text preprocessing and TF-IDF vectorization, turning unstructured feedback into ranked, actionable insights.

Tech Stack: *Python, NLP, TF-IDF*

Diabetes Risk Prediction

- Predicted diabetes risk from patient health records by running a complete ML workflow: data cleaning, feature engineering, model training, and evaluation with precision, recall, and ROC-AUC.

Tech Stack: *Python, Scikit-learn*

ML Model Deployment (REST API)

- Served real-time predictions from a trained machine learning model by containerizing a FastAPI/Flask REST API with Docker, demonstrating a production-style deployment workflow.

Tech Stack: *FastAPI, Flask, Docker*

Business Performance Dashboard

- Tracked revenue, profit, and core business KPIs in interactive Power BI dashboards by modeling data with SQL and Python, giving stakeholders a single self-serve view of performance.

Tech Stack: *Power BI, SQL, Python*

EDUCATION

B.Sc. in Computer Science

2022 – 2026

Future University in Egypt — Cairo, Egypt

- Specialization: Data Science, Machine Learning, and Artificial Intelligence.

EXPERIENCE & ACHIEVEMENTS

RoboCup Junior — Rescue Robotics (International Competition)

Germany | 2024

- **Won 2nd place in Egypt and 1st and 3rd place in the European rounds** by building autonomous rescue robots in Python with OpenCV and NumPy.
- Improved real-time victim-detection performance by optimizing computer-vision pipelines under strict on-board compute constraints.
- Mentored new team members and presented solutions at international workshops, strengthening team output across competition seasons.

Data Science & AI Trainee

Instant | 2026

- Prepared modeling-ready datasets by performing data cleaning, preprocessing, and feature engineering with Pandas and NumPy.
- Trained and fine-tuned machine learning and deep learning models, and evaluated them with precision, recall, F1-score, and ROC-AUC to select the best performers.
- Built NLP models for sentiment extraction and delivered interactive dashboards that communicated predictions for business and healthcare use cases.

Backend Developer — Academic Projects

- Shipped full-stack web applications with ASP.NET Core MVC and C#, exposing ML/AI model outputs through RESTful APIs.
- Reduced query response times by optimizing database access across MongoDB and SQL Server.

CERTIFICATIONS & TRAINING

- Data Science & AI Training — Python, Pandas, NumPy, Scikit-learn, TensorFlow, Keras (Instant, 2026)
- NVIDIA Deep Learning AI Fundamentals — Deep Learning, CNNs, GPU Acceleration (2026)
- Red Hat NLP & AI Solutions — Natural Language Processing, Text Mining, AI Applications (2025)
- Power BI for Data Analysis — Dashboards, Data Visualization, Business Intelligence (2025)
- Data Analysis with Python — Data Cleaning, EDA, Visualization (2025)
- Introduction to AI & Machine Learning — Udemy (2024)
- Python Programming (50 Hours) — Impact Platform, Zewail University (2023)

LANGUAGES

Arabic: Native | **English:** Professional Working Proficiency